This form documents the artifacts associated with the article (i.e., the data and code supporting the computational findings) and describes how to reproduce the findings.

Part 1: Data

☐ This paper does not involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).

☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Abstract

The paper uses two publicly available datasets and one simulation study. (1) The MovieLens 1M dataset provides user–movie ratings along with user and movie covariates; preprocessed data files are included in the repository. The data are publicly available online. (2) The Bixi bike-sharing dataset provides spatiotemporal trip counts; it is accessed via the BKTR R package. (3) Simulation data is generated by the provided scripts with fixed random seeds.

Availability

☒ Data **are** publicly available.
☐ Data **cannot be made** publicly available.

If the data are publicly available, see the *Publicly available data* section. Otherwise, see the *Non-publicly available data* section, below.

Publicly available data

☒ Data are available online at:

MovieLens 1M: https://grouplens.org/datasets/movielens/1m/

Bixi data: Accessed via the BKTR R package (`BixiData$new()`), available on CRAN at https://cran.r-project.org/package=BKTR

☒ Data are available as part of the paper's supplementary material.

The preprocessed MovieLens 1M data files are included in the repository under `data/movielens/raw/`. The code to obtain the Bixi data is also available in the repository under `code/bixi/1_generate_train_test_splits.R`.

☐ Data are publicly available by request, following the process described here:

☐ Data are available through some other mechanism, described here:

Non-publicly available data

Description

File format(s)

☐ CSV or other plain text.
☒ Software-specific binary format (.Rda, Python pickle, etc.): `.dat`, `.Rdata` and `.rds`
☐ Standardized binary format (e.g., netCDF, HDF5, etc.):
☐ Other (please specify):

Data dictionary

☒ Provided by authors in the following file(s): `data/README.md`
☐ Data file(s) is (are) self-describing (e.g., netCDF, HDF5, Zarr files)
☐ Available at the following URL:

Additional Information (optional)

The MovieLens license information: https://files.grouplens.org/datasets/movielens/ml-1m-README.txt Citation: F. Maxwell Harper and Joseph A. Konstan. 2015. The MovieLens Datasets: History and Context. ACM Transactions on Interactive Intelligent Systems (TiiS) 5, 4, Article 19 (December 2015), 19 pages. DOI=http://dx.doi.org/10.1145/2827872

The Bixi data is distributed as part of the BKTR R package. https://cran.r-project.org/package=BKTR

Part 2: Code

Abstract

The code reproduces all computational results in the paper: a simulation study comparing IMR against competing methods (SoftImpute, MCCI) across multiple settings, and two real-data applications (Bixi bike-sharing, MovieLens 1M). The proposed method is implemented in the R package `IMR` (provided as supplementary material and available on GitHub). The repository for the reproducible code is https://github.com/khaledfouda/imr-reproducible-code and for the R package is https://github.com/khaledfouda/IMR

Description

Code format(s)

☒ Script files

- ☒ R
- ☒ Python
- ☐ Julia
- ☐ MATLAB
- ☐ Other:

☒ Package

- ☒ R
- ☐ Python

- ☐ Julia
- ☐ MATLAB toolbox
- ☐ Other:

☐ Reproducible report

- ☐ Quarto or R Markdown
- ☐ Jupyter notebook
- ☐ Other:

☒ Shell script
☒ Other (please specify): Dockerfile for TensorFlow 1 virtual environment

Supporting software requirements

Primary software used

- R (>= 4.1), and one of the following:
- Python (>= 3.10): for the PyTorch GLocal-K implementation or
- Python 3.7 with TensorFlow 1.15.5 and Docker: for the original GLocal-K implementation (run inside a Docker container)

Libraries and dependencies used by the code R packages:

- `IMR` (0.1.2)
- `BKTR` (0.2.0)
- `softImpute` (1.4.3)
- `doParallel` (1.0.17)
- `parallel` (4.6.0)
- `tidyverse` (2.0.0)
- `dplyr` (1.2.1)
- `tidyr` (1.3.2)
- `magrittr` (2.0.5)
- `data.table` (1.18.4)
- `lubridate` (1.9.5)
- `Matrix` (1.7.5)
- `scales` (1.4.0)
- `kableExtra` (1.4.0)
- `ggh4x` (0.3.1)
- `RSSthemes` (1.0.0)
- `bench` (1.1.4)

Python packages (for GLocal-K only):

- `torch` (2.12.1) or `tensorflow` (1.15.5)
- `numpy` (2.5.0)
- `scipy` (1.18.0)
- `pandas` (3.0.3)

Full installation instructions are provided in `README.md`.

Supporting system/hardware requirements (optional)

No GPU is required. All computations run on CPU.

Parallelization used

- ☐ No parallel code used
- ☒ Multi-core parallelization on a single machine/node
  - Number of cores used: 9 but configurable and could run on a single core; we use `doParallel` for cross-validation.
- ☐ Multi-machine/multi-node parallelization
  - Number of nodes and cores used:

License

- ☐ MIT License (default)
- ☐ BSD
- ☒ GPL v3.0
- ☐ Creative Commons
- ☐ Other: (please specify)

Additional information (optional)

Part 3: Reproducibility workflow

Scope

The provided workflow reproduces:

- ☐ Any numbers provided in text in the paper
- ☒ The computational method(s) presented in the paper (i.e., code is provided that implements the method(s))
- ☒ All tables and figures in the paper
- ☐ Selected tables and figures in the paper, as explained and justified below:

Workflow

Location

The workflow is available:

- ☒ As part of the paper's supplementary material.
- ☒ In this Git repository: https://github.com/khaledfouda/imr-reproducible-code
- ☐ Other (please specify):

Format(s)

- ☐ Single master code file
- ☒ Wrapper (shell) script(s)
- ☐ Self-contained Quarto, R Markdown file, Jupyter notebook, or other literate programming approach
- ☒ Text file (e.g., a readme-style file) that documents workflow
- ☐ Makefile
- ☐ Other (more detail in *Instructions* below)

Instructions

Full instructions are provided in the repository's `README.md`. In summary:

1. Install R (>= 4.1) and the required R packages (listed in `README.md`).
2. Install Python (>= 3.10) and the required Python packages (`torch`, `numpy`, `scipy`, `pandas`) for the GLocal-K model.
3. Install the `IMR` package from the provided zip file or from GitHub.
4. From the repository root directory, run the master script:

```
chmod +x run_all.sh
./run_all.sh
```

This sequentially executes:

- **Simulation study:** `code/simulation/1_simulations.R` and `code/simulation/2_tables_plots.R`.
- **Bixi application:** `code/bixi/1_generate_train_test_splits.R`, `code/bixi/2_fit_BKTR.R`, `code/bixi/3_fit_IMR.R`, and `code/bixi/4_generate_results_table.R`.
- **MovieLens application:** `code/movielens/1_fit_MovieLens_models.R`, `code/movielens/2_1_prepare_python_da` `code/movielens/2_2_GlocalK_torch.py`, and `code/movielens/3_generate_results_table.R`.

Individual analyses can also be run independently; see the README files in each `code/` subdirectory.

Note: The `run_all.sh` script uses a non-official PyTorch implementation of GLocal-K. To use the original TensorFlow 1 implementation (which matches the reported results), Docker is required. See `code/movielens/README.md` for Docker build and run instructions.

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:

☐ < 1 minute
☐ 1-10 minutes
☐ 10-60 minutes
☐ 1-8 hours
☒ > 8 hours

To run the full analysis on an Apple M4 CPU: - The simulation study takes approximately 2–3 days (500 replicates × 4 dimension settings). - The Bixi BKTR model fitting takes approximately 4 days (250 training datasets, ~22 minutes each). - The MovieLens application takes approximately 8 hours for all models and cross-validation.

However, the time could be significantly reduced by reducing the number of replications. This can be achieved by modifying the relevant variables within the `helper.R` file in each code subdirectory.

Additional information (optional)

The `IMR` R package includes a vignette that demonstrates the method on example data. The vignette can be accessed after installation via `vignette("IMR")` in R.

Notes (optional)